

ASISH BAIDYA

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PROFILE SUMMARY

A final-year Power Engineering undergraduate at Jadavpur University with hands-on experience across the **data wrangling and engineering** to **predictive modelling** and **ML pipeline development**. Proven ability to build **scalable data pipelines** and deploy end-to-end machine learning solutions on real-world industrial and safety-critical datasets. Experienced in collaborating on research projects bridging **NLP, LLMs**, and sensor analytics, with a strong passion for **uncovering insights** that drive decision-making. Grounded in core engineering fundamentals including **IC engine thermodynamics** and **heat transfer**. eager to contribute fresh perspectives to product, engineering, and business teams solving high-impact data problems.

EDUCATION

Jadavpur University

B.Tech in Power Engineering | SGPA: 7.10

Kolkata, West Bengal

Aug 2023 – May 2027

Kalyan Nagar Vidyapith for Boys

Higher Secondary: 67.2% | Secondary: 81.14%

West Bengal

TECHNICAL SKILLS

Programming Languages: Python, C, MySQL

Core Concepts: Data Structures & Algorithms, Machine Learning, Deep Learning, Natural Language Processing (NLP), Retrieval-Augmented Generation (RAG), MLOps

Frameworks & Libraries: PyTorch, FastAPI

Tools & Platforms: Docker, Git, Power BI, Excel

EXPERIENCE

Research Intern — LLM-Assisted Crash Narrative Analysis

IIT Patna | Under Prof. Anilkumar Bacchu

- Integrating **LLM-assisted NLP** into an eDAR road safety framework to extract structured insights from unstructured accident reports, building **data pipelines** for automated pattern recognition in **safety-critical** datasets.
- Applying **statistical and analytical methods** for root-cause classification at high-risk road segments — transferable to **engine fleet performance** diagnostics and maintenance recommendation systems.

Student Researcher — Wireless Sensing Network

Ongoing

Under Prof. Gargi Konar, Jadavpur University Student Researcher — *Wireless Sensing Network* | GitHub

- Building an **ML pipeline** on real-time multi-sensor **accelerometer/telemetry data** (ATmega324PA + ZigBee) to predict surface roughness for autonomous drone landing — **algorithm development** on **industrial sensor data**.

PROJECTS

PaperTalks — Document AI & Recommendation System | | GitHub

2024

- Architected a scalable **RAG pipeline** over large PDF corpora using **FAISS vector embeddings** and LangChain orchestration, achieving **95%+ semantic extraction accuracy** on highly **domain-specific vocabulary**.
- Optimised chunking with RecursiveCharacterTextSplitter, cutting query response latency by **40%** while preserving cross-chunk context — demonstrates rigorous **algorithm design and evaluation**.
- Implemented **automated MCQ-based recommendation** engine and multi-PDF querying via spaCy NLP, simulating enterprise-scale **document intelligence workflows** applicable to **engine maintenance** records.

Fitness Tracker — Barbell Exercise Classifier | | GitHub

2024

- Built an end-to-end **ML pipeline** on raw **accelerometer/gyroscope sensor data**; benchmarked Naive Bayes, SVM, Random Forest, and Neural Networks, achieving **99.8% classification accuracy** — directly mirrors **sensor data** monitoring.
- Engineered time-frequency features using PCA, Butterworth filtering, and IQR-based outlier detection for robust preprocessing of **large-scale industrial sensor data**.
- Designed experiments**, conducted **thorough evaluations** across model variants — workflow directly aligned with GE Aerospace analytics responsibilities.

Adaptive English-to-Bengali Summarizer | | GitHub

- Engineered a two-pass **critic-in-the-loop LLM pipeline** using BART enforcing domain-specific constraints on readability and completeness, with a **feedback-driven adaptation** mechanism for continuous model refinement.
- Integrated a fine-tuned **cross-lingual translation model**, It will extend NLP accessibility to a 15M+ speaker language group — showcasing ability to build and evaluate end-to-end **NLP analytical solutions**.

ACHIEVEMENTS & CERTIFICATIONS

Meta OpenEnv AI Hackathon, Bengaluru — Top Finalist

Apr 2026

- Engineered a **multi-agent RL simulation** for smart grid energy dispatch (**industrial domain**) using OpenEnv-compliant distributed environment; implemented real-time reward modelling and agent-based market interactions under competitive time constraints.

Certifications: Career Essentials in Data Analysis — Microsoft & LinkedIn | Power BI 7-Day Micro-Course